

3.3 – 3.4 Worksheet

Determine whether the following series converge or diverge. Use any test that is helpful.

1. $\sum_{k=1}^{\infty} \frac{7}{3k}$

2. $\sum_{k=1}^{\infty} \frac{4k^3 + 2k}{2k^3 - 1}$

3. $\sum_{i=1}^{\infty} \operatorname{sech}^2(i)$ (*Hint: it may be helpful to consider a graph of $y = \operatorname{sech}^2(x)$).*

4. $\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$